

Nomenclature and Principles of **Disease**

- **Characteristics, classification & incidence of disease**

What is disease?

- A condition in which the presence of **an abnormality of the body** causes a loss of normal health
- Disease = illness = ill health
- Limits of normality

Characteristics of disease

- **Aetiology** -> cause
- **Pathogenesis** -> mechanism of development of disease
- **Pathologic & clinical features**-> functional & structural alterations
- **Pathognomonic** abnormalities
 - ✓ Restricted to a single disease
 - ✓ **Diagnosis is certain in their presence** e.g.
Reed-Sternberg cells in Hodgkin's lymphoma

Characteristics of disease ctd

- Lesion is the structural & functional abnormality responsible for ill health
 - It could be visible or biochemical
- **Complications & sequelae**: systemic, 2^o or remote effects
- **Prognosis** forecasts the known or likely course of the disease in terms of cure, remission or fate of the patient
 - In cancer -> 5 yrs survival rate

Causes of disease

- Introduction
 - ✓ Host & environment **interaction**
 - ✓ Predictable e.g. **Injury**
 - ✓ Probable e.g. **Infection**
 - ✓ **Genetic** predisposition
 - ✓ Some diseases predispose to other e.g. AIDS, liver cirrhosis

Causes of disease ctd

- ✓ Multi factorial eg. IBD
- ✓ Unknown (idiopathic, essential, cryptogenic or 1^o)
- ✓ Features that suggest **causal association**
 - ❖ Genetic: family, gender, race, blood group, HLA
 - ❖ Environment: occupation & geography

Specific causes

➤ Acquired or genetic

Acquired

1. **Microorganisms** e.g. bacteria, virus,...
2. **Chemicals** e.g. drugs, acids,...
3. **Physical agents** e.g. radiation, ...
4. **Nutritional** (under or over) e.g. protein energy, vitamin, ...
5. **Immunological** e.g. Immunodeficiency, autoimmunity, hypersensitivity

Specific causes ctd

6. **Metabolic** e.g. Diabetes mellitus

7. **Vascular** e.g. Atherosclerosis

8. **Psychogenic** e.g. Mental disease, addiction, ...

❖ **Iatrogenic diseases** are attributable to physicians or health professionals action & **follows investigation or treatment**

Nomenclature of disease

1. Primary & secondary

- Causation => 1^o (unknown) or 2^o (known)
e.g. Hypertension could be 1^o or 2^o
- Stages of a disease => 1^o or 2^o effects
e.g. 1^o cancer -> disseminate-> 2^o

2. *Acute & chronic* : for *dynamics* of a disease

- Acute - rapid onset
- Chronic - insidious onset & prolonged course
- Subacute in between the above two

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3. **Benign & malignant** : for likely outcome of a disease

- Benign - very rarely lethal or fatal
- Malignant - commonly lethal

4. Prefixes

- Hyper - Excess e.g. Hyperthyroidism
- Hypo - Deficiency e.g. Hypothyroidism
- Meta – a change from one state to another
e.g. metaplasia

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4. Suffixes

- itis: inflammatory process e.g. gastritis
- oma: tumor e.g. carcinoma
- osis: state or condition e.g. osteoporosis
- oid: bearing a resemblance to e.g. rheumatoid disease
- plasia: disorder of growth e.g. hyperplasia
- opathy: abnormal state lacking specific characteristic e.g. lymphadenopathy

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- **Eponymous**: named *after a person or place* associated with it e.g. Hodgkin lymphoma
- **Syndromes**: an *aggregate of signs & symptoms* or a *combination of lesions* without which the disease cannot be recognized e.g. Cushing, Down, ...